

THIRANEX INTERNSHIP

Task 3 – Exploratory Data Analysis (EDA)

Netflix Movies and TV Shows Dataset

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Tools: Python, Pandas, NumPy, Matplotlib, Seaborn, Google Colab

A practical Exploratory Data Analysis project focused on understanding Netflix content through data cleaning, statistical analysis, visualization, and identification of meaningful patterns.

1. Introduction

Exploratory Data Analysis (EDA) is an important step in the data science process. It helps understand the structure of a dataset, identify missing values and duplicates, summarize important variables, and discover patterns through visualizations.

In this project, the Netflix Movies and TV Shows dataset is analyzed to understand the types of content available, ratings, countries, genres, release years, and movie durations.

2. Project Objectives

- Explore the Netflix dataset and understand its structure.
- Identify and handle missing values.
- Remove duplicate records.
- Generate statistical summaries.
- Compare Movies and TV Shows.
- Analyze countries and genres.
- Study ratings and release-year trends.
- Analyze movie duration.
- Examine numerical correlations.

3. Dataset Description

The dataset contains information about Netflix Movies and TV Shows. Important fields include show ID, type, title, director, cast, country, date added, release year, rating, duration, listed genres, and description.

4. Technologies Used

Python was used as the main programming language. Pandas and NumPy were used for data processing and numerical operations. Matplotlib and Seaborn were used to create visualizations. Google Colab was used as the development environment.

5. Data Cleaning

The dataset was copied before preprocessing. Missing values in text columns were replaced with 'Unknown', while numerical missing values were handled using the median. Duplicate rows were removed to improve data quality.

6. Exploratory Analysis

Analysis	Purpose
Movies vs TV Shows	Compare the two main content types.
Top Countries	Identify countries producing the most content.
Release Year	Study content distribution across years.
Genres	Find the most common Netflix genres.

Ratings	Understand the distribution of content ratings.
Movie Duration	Analyze the typical duration of movies.
Correlation	Examine relationships between numerical variables.

7. Visualizations

The project includes visualizations for content type, top content-producing countries, release-year distribution, popular genres, ratings, movie duration, and numerical correlations. These charts make it easier to understand trends and compare different categories.

8. Key Findings

- Netflix contains both Movies and TV Shows, with Movies forming a large share of the catalog.
- Content comes from many countries, showing the international nature of the platform.
- A smaller group of genres appears more frequently than others.
- Netflix content can be studied effectively through release year and rating distributions.
- Movie duration varies considerably across titles.
- Visual analysis provides a clear way to identify major patterns in the dataset.

9. Conclusion

The Exploratory Data Analysis of the Netflix Movies and TV Shows dataset demonstrates how data cleaning, statistical summaries, and visualization can be combined to understand a real-world dataset. The analysis provides useful insights into Netflix's content library and establishes a strong foundation for further data science or machine learning work.

10. Future Scope

The project can be extended by building recommendation systems, predicting content trends, performing advanced time-series analysis, or applying machine learning techniques to Netflix data.

Project Status: Completed

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